

Follow-Up Is Not Discovery: What a Record of Prior Audit Findings Can and Cannot Tell You

Abstract

Audit findings are both outcomes of prior work and inputs to future audit planning. When prior findings change subsequent verification, the finding record combines information about underlying conditions with information about where auditors looked. We characterize what that record can reveal. Three programs whose underlying condition is persisting, unchanged, or improving can produce the same finding history, so the record alone does not identify the underlying association. It does bound it: if follow-up can only raise detection, the observed finding risk ratio is an upper bound on the underlying-condition risk ratio, and a defensible ceiling on the follow-up detection multiplier makes the bound two-sided. Reference observation under a common measurement rule, or other information about detection, can make the bounds operational. Prior findings are natural routing information for follow-up; what they reveal about discovery cannot be inferred from the finding record alone. In Federal Audit Clearinghouse data, prior findings predict both recurrence and materially more nonrepeat findings among programs audited as major in adjacent years, while a recurrence-share statistic ranges from 61.2 to 90.9 percent with the allocation of mixed outcomes alone. A risk score validated against recorded findings can improve while carrying no information about underlying conditions.

Keywords: audit findings; follow-up; detection; risk assessment; partial identification; Single Audit; predictive validation.

Data: all data used in this study are public Federal Audit Clearinghouse bulk files; Appendix C describes the acquisition and the reproduction path.

1 INTRODUCTION

Audit findings are used twice. They are outputs of completed work, and they become inputs to the next audit plan. A prior finding can change what the auditor tests, how much is tested, and which known conditions receive follow-up. The next finding record is therefore produced partly by what problems exist and partly by where the audit chose to look.

This paper asks what a record of prior findings can tell us about future underlying problems when the prior findings themselves change future observation. The answer depends on which of two jobs the record is being asked to do. Follow-up asks whether a known condition remains. Discovery asks where conditions not already represented in the finding history exist. A prior finding is close to ideal routing information for follow-up: it identifies a known condition that deserves revisiting, and returning there is what a risk-based audit is for. It is not sufficient verification that the condition remains; the follow-up work generates that evidence. Discovery is different. Prior findings can contain information about discovery, because some entities are persistently susceptible across the board and one underlying weakness can affect several kinds of condition. But evidence generated by returning to known conditions does not by itself identify where unreported conditions lie, and the finding record alone does not identify how much of that relationship reflects underlying conditions and how much reflects the observation process created by the prior findings themselves, because the places without findings are precisely where absence, lack of observation, and failed detection leave the same recorded outcome.

The point can be made exact with three quantities. Write F_t for whether a finding is recorded for a program this year, C_t for whether the underlying condition actually exists this year, and f for whether a finding was recorded in the prior period, so $f = 1$ after a finding year and $f = 0$ after a clean year. A finding is recorded only if the condition exists and the audit detects it, so the chance of a finding this year, given last year's record, factors into two pieces:

$$P(F_t = 1 \mid F_{t-1} = f) = P(C_t = 1 \mid F_{t-1} = f) \times P(F_t = 1 \mid C_t = 1, F_{t-1} = f). \quad (1)$$

Give the three pieces short names. q_f is the observed probability of a current finding after prior history f : what the record shows. r_f is the probability the underlying condition is actually present after history f : what a planner wants to know. d_f is the probability the audit detects the condition when it is present, after history f : how hard and where the audit looked. Equation (1) then reads

$$q_f = r_f d_f. \quad (2)$$

In audit terms: what appears in the finding record depends on two things, how often the problem is actually there and how often the audit catches it when it is there. A prior finding can move either one. Remediation changes r . Follow-up changes d . The record shows only q . Figure 1 draws the sequence, together with the feedback arrow that gives a prior finding its second use: this year's record enters next year's risk assessment, follow-up plan and scope, and so changes next year's d . When that arrow is active, a difference in q between history groups is a difference in r , a difference in d , or both, and the record does not say which. All three quantities compare groups defined by prior recorded history; they are not before-and-after effects of remediation or follow-up on a given program, which are the audit mechanisms that motivate them.

One assumption is maintained throughout: a recorded finding reflects a real condition. If findings can be recorded where no condition exists, at rate a_f after history f , the recorded rate becomes $q_f = r_f d_f + (1 - r_f) a_f$. Equation (2) and every bound built on it hold under no false positives, or under a modeled restriction on a_f carried through the algebra (Appendix A).

Verification bias, imperfect detection, endogenous inspection and selective labels are established problems (Ransohoff and Feinstein, 1978; Begg and Greenes, 1983; MacKenzie et al., 2002; Heckman, 1978; Lakkaraju et al., 2017; Waymire et al., 2018). Audit research and practice use prior findings both to measure risk and to determine subsequent verification. The audit-finding studies most directly related to this paper document recurrence and predictive association, and recognize that a reported finding depends on the condition and its detection (Keating et al., 2005; Petrovits et al., 2011; Rice and Weber, 2012; Waymire et al., 2018), but do not separately identify

the latent-risk and observation components of the finding history. This paper shows that the two uses make the resulting finding history endogenous to the audit design, derives what remains identified, and shows how audit design can recover the missing information. Four contributions follow. *Identification:* Section 2 shows exactly why a finding history cannot, on its own, separate persistence of underlying problems from persistence of observation, by building three programs with the same record whose condition persists, is unchanged, or has been halved. *Positive bounds:* Section 3 shows that if an existing condition is at least as likely to be detected after a prior finding as after a clean history, the observed finding risk ratio is an upper bound on the underlying-condition risk ratio, and that a defensible ceiling on the follow-up detection multiplier makes the bound two-sided; Section 8 sets out how an organization can learn or bound that multiplier. *Measurement:* Section 5 distinguishes repeat-only, new-only and mixed outcomes and shows why a recurrence-share statistic can range from 61.2 to 90.9 percent without any change in the data. *Empirical illustration:* in Federal Audit Clearinghouse data, prior findings predict both recurrence and materially more findings the auditor did not classify as repeats, among programs audited as major programs in both adjacent years, while the record itself cannot identify why.

The Single Audit is the illustration because history-dependent observation is written into its rules rather than inferred from auditor behavior; Section 4 sets those rules out before the data are used. The argument itself is general: regulatory enforcement, quality assurance, clinical surveillance and tax examination all produce records with the same property. Proofs are in Appendix A.

2 THREE PROGRAMS, ONE FINDING HISTORY

Consider three programs. In each, the finding record shows the pattern auditors recognize: a program with no finding last year has about a 0.05 chance of a recorded finding this year, and a program with a finding last year has about a 0.20 chance. In the notation of equation (2), $q_0 = 0.05$ and $q_1 = 0.20$. Recorded findings are four times as likely after a finding year as after a clean year. Any audit shop would read that as a strong predictor of another recorded finding, and would

reasonably plan accordingly.

Underneath, the three programs have nothing in common. Table 1 sets them side by side.

In the first, the condition really did persist. The chance the deficiency is present rises from $r_0 = 0.100$ after a clean year to $r_1 = 0.400$ after a finding year, and the audit catches half of what is there in both cases, $d_0 = d_1 = 0.500$. Multiply: 0.100 times 0.500 is 0.05, and 0.400 times 0.500 is 0.20. Equal detection halves the absolute rates and their difference (a recorded gap of 0.15 against an underlying gap of 0.30) and leaves the risk ratio exactly as it is underneath.

In the second, nothing about the condition changed at all. The deficiency is present $r_0 = r_1 = 0.200$ of the time regardless of last year. What changed is the looking. After a clean year the auditor catches $d_0 = 0.25$ of what is there; after a finding year, under a follow-up requirement, the auditor catches all of it, $d_1 = 1.0$. Multiply: 0.200 times 0.25 is 0.05, and 0.200 times 1.0 is 0.20. Identical record. The entire fourfold elevation is the follow-up, and the prior finding carries no information about whether the condition is there this year.

The third case is the one worth sitting with. Here remediation is stipulated to halve the underlying rate: the chance the deficiency is present *falls* from $r_0 = 0.400$ after a clean year to $r_1 = 0.200$ after a finding year. But the follow-up is thorough. Detection went from $d_0 = 0.125$ to $d_1 = 1.0$, a factor of 8. Multiply again and you get 0.05 and 0.20, the same two numbers. In the record, programs coming off a finding year look four times worse than programs coming off a clean year, and in this third case it is the finding-year programs whose underlying rate is half as high. This column is a constructed world compatible with successful remediation. It is not evidence that remediation caused any decline; it shows that the record cannot rule the possibility out.

The arithmetic is exact, and the three cases are three points on a continuum along which the observed record is constant.

The observed ratio is a product

The comparison auditors actually make is a ratio: how much more often findings are recorded after a prior finding than after a clean year. Three ratios are needed to say what that comparison

contains.

$R_{\text{obs}} = q_1/q_0$ is the *observed finding risk ratio*: how much more often findings are recorded after a prior finding. In Table 1 it is 4 in every column, and it is the only one of the three the record reports.

$R_{\text{lat}} = r_1/r_0$ is the *underlying-condition risk ratio*: how much more often the problem itself exists after a prior finding. It is what a planner would like to know.

$M = d_1/d_0$ is the *history-conditioned detection multiplier*: how much more likely an existing problem is to be detected in the prior-finding group than in the clean-history group. Follow-up is the audit mechanism that motivates it, and the paper calls it the follow-up detection multiplier for short. It is a ratio of two detection probabilities, not a measure of hours, sample size or effort.

All three ratios compare groups defined by prior recorded history. None of them is a before-and-after effect of remediation or of follow-up on a given program.

Dividing $q_1 = r_1 d_1$ by $q_0 = r_0 d_0$ gives

$$R_{\text{obs}} = R_{\text{lat}} \times M. \quad (3)$$

The ratio of recorded finding rates is the product of the ratio of underlying-condition rates and the ratio of detection probabilities. The fourfold finding ratio in Table 1 can come from a fourfold ratio in underlying problems, a fourfold ratio in detection, or any combination whose product is four. The table's last three rows show the three cases: in Persistence, $R_{\text{lat}} = 4$ and $M = 1$; in Follow-up only, $R_{\text{lat}} = 1$ and $M = 4$; in Remediation, $R_{\text{lat}} = 0.5$ and $M = 8$. Each multiplies to $R_{\text{obs}} = 4$. A fourfold finding ratio does not by itself establish fourfold underlying risk, and it does not establish that follow-up is working either. More years of the same kind of data do not help, because every additional year is generated by the same product.

Successful remediation can raise recorded findings

One corollary has a direct operational consequence. Suppose the underlying condition is *less* common in the prior-finding group, $r_1 < r_0$, as successful remediation would make it. Recorded findings still rise after a finding whenever

$$M > r_0/r_1, \tag{4}$$

that is, whenever the follow-up detection multiplier exceeds the factor by which the problem was reduced. If remediation halves the true problem rate but follow-up makes detection more than twice as likely, the finding record rises even though the underlying condition improves. In the Remediation column, $r_0/r_1 = 2$ and $M = 8$, and eight exceeds two. An organization that judges its corrective-action process by comparing recorded findings across history groups is using a comparison that can run the wrong way precisely when the corrective action works and the follow-up is diligent (Proposition A.4).

3 WHAT THE RECORD IDENTIFIES, AND WHAT IT ONLY BOUNDS

The finding record does not point-identify the underlying condition, but it does impose useful bounds. Three statements hold, needing in turn no additional assumption about detection, one assumption about the direction of follow-up, and one number about its size. All three rest on the no-false-positive assumption of Section 1.

With no additional assumption about detection: a floor and a wide range

Because detection cannot exceed 100 percent, the true problem rate in each history group must be at least as high as the recorded finding rate: $d_f \leq 1$ in equation (2) gives $r_f \geq q_f$. If five percent of clean-history program-years record a finding, at least five percent had the condition. Without

known complete detection, a recorded finding rate is a lower bound on the underlying condition rate, not an estimate of it.

The question a planner asks is about the ratio, and the floor alone leaves it wide. R_{lat} is largest when the prior-finding group's problem rate is as high as possible (one) and the clean group's is as low as the record allows (q_0), and smallest in the reverse case, so

$$R_{\text{lat}} \in [q_1, 1/q_0], \quad (5)$$

and both ends can actually occur (Proposition A.2). At the values in Table 1, $q_0 = 0.05$ and $q_1 = 0.20$, so R_{lat} can range from 0.20 to 20. The same fourfold recorded ratio is consistent with the underlying condition being five times less common, equally common, or much more common after a prior finding. Without information about detection, the record does not even fix the sign of the underlying association.

With one assumption: the observed ratio is a ceiling

Now assume that follow-up never makes the auditor look *less* hard at a program that had a finding than at one that did not: an existing problem is at least as likely to be detected after a prior finding as after a clean history,

$$d_1 \geq d_0, \quad \text{equivalently } M \geq 1.$$

Call this monotone detection. The direction is consistent with a follow-up regime, which directs more attention toward prior findings and never less, but monotone detection remains a substantive assumption about how follow-up affects the probability of detecting an existing condition, not a fact the regulation proves; it fails, for example, when a prior finding shifts attention toward one area and away from another, and it may hold for some kinds of condition and not others.

Under monotone detection, dividing equation (3) by $M \geq 1$ gives

$$R_{\text{lat}} \leq R_{\text{obs}}. \quad (6)$$

If follow-up can only increase detection, the recorded risk ratio cannot be smaller than the underlying risk ratio. In Table 1, the underlying ratio is at most 4 in every column, and the columns show it taking the values 4, 1 and 0.5. One consequence needs no further information: if the recorded ratio is below one, so that programs with a prior finding record *fewer* findings than programs without, the underlying association is genuinely negative, and no amount of differential follow-up can explain it away.

With one number: a floor on the underlying ratio

The constructive result comes from putting a ceiling on the follow-up detection multiplier. Write $\overline{M}$ for the largest value of M the organization is prepared to defend: the most that follow-up could plausibly raise the chance of detecting an existing problem. If $1 \leq M \leq \overline{M}$, dividing equation (3) by $M \leq \overline{M}$ gives

$$R_{\text{lat}} \geq R_{\text{obs}}/\overline{M}. \quad (7)$$

Even after giving follow-up the largest detection advantage the organization considers plausible, the underlying condition is still at least $R_{\text{obs}}/\overline{M}$ times as common after a prior finding. A recorded ratio of 4 and a defensible belief that follow-up can increase detection by no more than a factor of 2 implies an underlying risk ratio of at least 2. With equation (6), the record then places R_{lat} between 2 and 4: the inference survives, cut down by the part of the elevation the follow-up can account for. And whenever the recorded ratio exceeds $\overline{M}$, the underlying association is genuinely positive (Proposition A.3).

These bounds are sharp once the floor $r_f \geq q_f$ is added. Each detection probability lies between q_f and one, so the multiplier cannot exceed $1/q_0$, and the multipliers consistent with the record are $1 \leq M \leq \min\{\overline{M}, 1/q_0\}$. The identified set is

$$R_{\text{lat}} \in \left[\max\left\{q_1, \frac{R_{\text{obs}}}{\overline{M}}\right\}, R_{\text{obs}} \right], \quad (8)$$

and every value in it is produced by some admissible pair of detection probabilities (Proposi-

tion A.3(iii)). At the values of Table 1 with $\bar{M} = 2$ the set is $[2, 4]$.

The binding identification constraint on what an auditor can learn from a finding record is therefore not sample size or a more elaborate estimator; it is what can be said about detection, summarized in M . Sample size still governs precision. The propositions are population statements about q_0 and q_1 ; in data, R_{obs} is estimated, and each conclusion needs the matching confidence limit. A negative underlying association requires an upper confidence limit for R_{obs} below one. A positive association under a defended $\bar{M}$ requires a lower confidence limit for R_{obs} above $\bar{M}$. What can be said about detection is a question about audit procedures, answerable by the organization that performs them and not from the record; Section 8 sets out how.

More years of history do not remove the problem

A longer record does not solve this on its own. Take any recorded history of any length and any story about how likely the underlying problem was at each point, so long as it stays at or above the recorded rate. Setting detection at each point to whatever value reconciles the two reproduces the entire recorded history exactly (Proposition A.5). Adding years adds observations of the product $q = rd$, not of either factor. Restricted models of the underlying condition can nevertheless be identifiable when substantive restrictions constrain how the condition evolves and how detection responds to the record (Allman et al., 2009); identification then comes from those restrictions, not from record length alone. This paper does not estimate such a model. Its interest is what can be said before those restrictions are imposed, and what audit design can add.

4 THE SINGLE AUDIT AS A HISTORY-DEPENDENT OBSERVATION REGIME

The empirical illustration comes from the Single Audit, and its terms need defining before the data are read.

A Single Audit is an organization-wide audit required of nonfederal entities, such as states,

local governments, universities and nonprofits, that expend federal awards above a regulatory threshold. It combines a financial statement audit with compliance testing of the entity's federal programs, and its structured results are published by the Federal Audit Clearinghouse.

Compliance testing is not spread evenly across every program. The auditor designates certain federal programs as *major programs* and performs the required major-program compliance audit on them. Programs not selected as major do not receive that full program audit solely by virtue of program status, although other work, including required follow-up of prior findings, may still occur. The selection is risk based. Programs whose federal expenditures exceed the Type A threshold are *Type A*; the rest are Type B. The threshold is tiered by the entity's total federal expenditures: over the sample period it was \$750,000 for entities expending up to \$25 million and higher in tiers above that (2 CFR 200.518(b)(1)). A Type A program that meets a list of low-risk criteria may be left out of detailed testing in a given year, and one that fails the criteria must be tested as major. Prior findings are among the criteria: a material weakness, a modified opinion on the program, or questioned costs above five percent of the program's expenditures in the most recent audit period disqualify a Type A program from low-risk treatment.¹ The requirements tested for a major program are those the Office of Management and Budget's annual *Compliance Supplement* designates as applicable to it.

Separately, and regardless of program selection, the auditor must follow up on prior audit findings, whether or not the program with the prior finding is selected as major again. In the governing regulation, prior findings raise assessed program risk (2 CFR 200.519(b)(2)); they can disqualify a Type A program from low-risk treatment (200.518(c)(1)); and follow-up is mandatory "regardless of whether a prior audit finding relates to a major program in the current year" (200.514(e)). None of this is a defect; the follow-up requirement exists because findings left uncorrected are a documented problem in this program (U.S. Government Accountability Office, 2002).

The setting is useful for one reason. In most inspection systems the feedback from past results

¹Not every prior finding does this. A significant deficiency that is not a material weakness, or questioned costs below the five-percent line, leaves a Type A program eligible for low-risk treatment. The paper does not assume that every prior finding makes major-program coverage more likely, and its sample construction conditions on major-program coverage in both years.

to present attention has to be inferred from inspector behavior. Here it is written into the rules, through two channels: a coverage channel, in which a prior finding can change whether a program is tested in depth at all, and a condition-specific channel, in which a prior finding must be followed up wherever it sits. Restricting the sample to programs audited as major in both years conditions on the major-program-selection channel: every comparison is among programs selected in both years. That conditioning does not make the selection ignorable, and it does not hold audit intensity or condition-specific observation fixed: the follow-up channel is mandated inside the restricted sample, so two such programs, one with a prior finding and one without, face different condition-specific attention by regulation even though both are audited in depth in both years.

5 FAC EVIDENCE: RECURRENCE, NEW FINDINGS, AND MIXED CASES

Prior work has established that Single Audit findings recur and that a reported finding depends jointly on the condition existing and on the auditor detecting it (Keating et al., 2005; Petrovits et al., 2011; Waymire et al., 2018; U.S. Government Accountability Office, 2002, 2024; Waddell, 2023). What is added is the decomposition of what follows a prior finding into recurrence, findings not classified as repeats, and mixed cases.

Data and sample

The data are the public Federal Audit Clearinghouse bulk files; the unit of analysis is the auditee–program–year, and the window is fiscal years 2016 through 2023, inside a single audit-threshold regime (Office of Management and Budget, 2024). Of 2,387,511 program-years, 1,600,898 have a resolved prior-year observation of the same program, and Sample A restricts further to the 206,926 in which the program was audited as a major program in both years: 39,067 with a prior-year finding and 167,859 without. Intervals are cluster bootstrap over 45,952 auditees. Appendix C gives the construction and the sample flow.

Write $J_t = 1$ when the program is audited as a major program in year t . Every Sample A quantity is conditional on $J_{t-1} = J_t = 1$: the comparisons are $P(F_t = 1 \mid F_{t-1} = f, J_{t-1} = J_t = 1)$ for $f = 0, 1$, and the empirical statements below apply to that selected population. Selection into it depends on the finding record (Section 4), so the conditioning defines the population described; it does not make selection ignorable for any wider population.

One construction decision follows from the paper’s own argument. In 366,506 program-years, 15.35 percent of the panel, the auditee filed in the prior year but this program does not appear in that filing. They are excluded rather than coded as “no prior finding”, because the absence of a record is not a clean result: they go on to record a finding 3.3 percent of the time, against 2.1 percent after a resolved clean prior year. “Repeat” is the auditor’s own coding; a validated historical link is the secondary definition (Appendix C). A finding is *new* when the auditor did not classify it as a repeat. “New” is shorthand for that classification: it does not establish that the condition was never recorded before or that it arose this year.

What follows a prior finding

Even among programs audited as major in both consecutive years, prior recorded findings predict two distinct parts of the future finding record: explicit recurrence, and additional findings not classified as repeats. Table 2 is the main result. For context, a program-year following a clean prior year records a finding 0.08 of the time and one following a prior-year finding records one 0.58 of the time, a difference of 0.50 [0.49, 0.51] and a ratio of 7.55. The content is in how that difference is made up.

A program-year lands in exactly one of four states: no finding (0); a repeat finding and nothing else (R); a new finding and nothing else (N); or both a repeat and a new finding (B). Write Δ for the difference in the probability of a state between program-years with and without a prior finding. Because the four states do not overlap and cover every possibility, the differences add:

$$\Delta_{\text{any finding}} = \Delta_R + \Delta_N + \Delta_B. \quad (9)$$

The categories can be added without deciding where mixed cases belong, because each mixed case is counted once, in B . In Sample A, $\Delta_R = +0.3073$, $\Delta_N = +0.0456$ and $\Delta_B = +0.1488$, adding to $+0.5017$. Every interval in the table excludes zero.

Most of the gap is repeat-involving, as an auditor would expect. The row that is easy to read past is the one about new findings. Programs whose *only* outcome was a new finding account for $+0.0456$, a ratio of 1.61, which looks like almost nothing. But “new only” does not answer the question a reader is asking, which is whether the program recorded *any* new finding. That quantity is

$$\text{any new} = N + B,$$

because a period that recorded both a repeat and something new did record something new. On that margin the difference is $+0.1945$ $[0.1870, 0.2021]$, from 0.08 to 0.27, a ratio of 3.59. The two ratios, 1.61 and 3.59, differ by a factor of 2.23 and come from the same table. The difference between them is definitional, not evidence about follow-up: “new only” excludes every period that also recorded a repeat, and “any new” includes them. Periods land in B for more than one reason, and these data do not separate them: thorough follow-up of a repeat can put a program there that would otherwise have shown only a new finding, and so can a program simply having several distinct deficiencies.

The cleanest version counts findings rather than flagging programs. Every individual finding is classified once, as a repeat or as new, so the difference in the expected number of findings splits with no allocation rule at all:

$$\underbrace{+1.0683}_{\text{total}} = \underbrace{+0.7441}_{\text{repeat}} + \underbrace{+0.3243}_{\text{new}}. \quad (10)$$

The unrounded values satisfy the identity exactly; the four-decimal values shown differ in the last digit by rounding. A program-year following a prior-year finding records about one more finding than one following a clean year, of which about seven tenths of a finding is a repeat and three tenths is new.

The record does not identify how much of these associations reflects true persistence of the underlying condition, stable differences between auditees, condition-specific follow-up, broader scrutiny of programs with a history, remediation combined with heavier detection, or mixtures of these. Each is consistent with Table 2. The FAC results illustrate the identification problem; they do not identify its mechanism, and nothing here should be cited as showing that observation feedback is the dominant explanation.

A recurrence share depends on how mixed outcomes are allocated

The obvious way to compress Table 2 into one number is to ask what share of the prior-finding association is recurrence. Every mixed period, 25.9 percent of all prior-finding outcomes, has to be assigned to one side first. Assign none of them to recurrence and the share is 61.2 percent. Assign all of them and it is 90.9 percent. The summary moves 29.7 percentage points, and nothing in the underlying observations changes (Figure 2). A two-category recurrence share is not uniquely defined until the treatment of mixed outcomes is specified, and it is not a causal attribution. For a fixed allocation fraction ζ it is a well-defined descriptive function of observable quantities. The general lesson: when outcome classes overlap, report mutually exclusive states or additive counts before reporting an attributable share, and if a share is reported anyway, state the convention and show the figure under the opposite one. The convention-free statement is equation (9): $0.5017 = 0.3073 + 0.0456 + 0.1488$.

A comparison the data do not support

A natural next question is whether prior findings predict problems in requirement areas that never had one. The Clearinghouse records the requirement category attached to every reported finding. It does not record the universe of requirements that applied to, and were tested in, each program-year; that universe is defined by the Compliance Supplement and is external to these data. A “first finding in a previously clean area” rate therefore has no defensible denominator, and without one the comparison goes wrong because the two history groups do not have the same opportunity

to produce the outcome. Take two requirement areas, each with a 0.20 chance of a finding this year, independent of each other and of anything before. A program with neither area previously flagged has a $1 - 0.80^2 = 0.36$ chance of a finding in an area new to it; a program with one area already flagged has only the other available, so its chance is 0.20. Nothing underneath differs. In a constructed twelve-area case in which nothing depends on history, the naive new-area risk ratio is 0.72: prior-finding programs look about thirty percent *less* likely to have a new kind of problem, from unequal opportunity alone. No denominator was invented, and the comparison is not made.

Robustness

Table 8 in Appendix C reports every specification. Restricting to transitions with stable Clearing-house provenance moves the any-finding gap from 0.5017 to 0.4885 and leaves any-new at 0.1949. Substituting the validated historical link for the auditor’s repeat flag moves the repeat-only gap to 0.2994 and the both-kinds gap to 0.1454, leaves any-new unchanged, and opens a fifth outcome state worth 0.0079 of the gap, given its own column there. A prespecified three-year history rule, under which “no prior finding” means three observed clean years, leaves 94,360 program-years and attenuates the gaps to 0.3674 for any finding and 0.1716 for any new. Across all five specifications the any-new margin moves least.

6 FOLLOW-UP AND DISCOVERY

Auditors already have the vocabulary for the distinction the results point at. Follow-up is returning to a previously identified problem. Discovery is learning where problems exist outside what has already been identified. Both are part of the work, and they are not the same job.

For follow-up, a record of prior findings is close to ideal routing information: it identifies the known conditions that deserve revisiting, and the regulation is right to require the visit. It is not sufficient verification that a condition remains; the follow-up procedures generate that evidence, and the Single Audit results are consistent with follow-up working as designed. Discovery con-

cerns conditions not already represented in the finding history. Prior findings can carry genuine information about it: some entities are durably more susceptible across the board, and one underlying weakness can produce several kinds of condition. What the record cannot say is how much of that apparent signal is susceptibility and how much is the observation process, because the places without findings are precisely where absence and non-detection cannot be separated, and because a prior finding can widen the search. A record built by looking where the record said to look contains information about the places that were looked at; how much it contains about everywhere else is the question it cannot answer about itself.

The distinction has a testable shape, because different audit responses to a prior finding leave different marks on the outcomes of Table 2. Table 3 works through three ways an auditor might let last year's finding change this year's work, with the underlying problems held identical and drawn independently each period, so nothing underneath persists at all. Every cell is the change in the prior-finding gap relative to a baseline in which prior findings change nothing. The regimes share their first-period histories, so the history groups and each program's previously flagged conditions and areas are the same in every regime; they share second-period conditions and detection draws program by program; and each regime changes detection only for the conditions it targets. An outcome outside a regime's target set is therefore identical, program by program, to its baseline value, which makes the zeros exact and cancels the opportunity-set artifact of Section 5. New findings are separated by whether they fall in a requirement area that already had one, because that is where the three responses part company.

Condition-specific follow-up. If the response is to re-test the control or re-examine the transactions behind the specific finding, repeat findings rise and nothing else moves; the two new-finding outcomes are exactly zero.

Category-level follow-up. If the response is to tighten work across the whole requirement area that produced the finding, repeats rise and so do new findings inside that area, while areas never flagged remain untouched.

Generalized scrutiny. If the response is general wariness, so that a program with a finding

simply gets a harder audit everywhere, all three outcomes move, including findings in requirement areas that never had one. That last column is the one to watch, because a finding in a previously clean area is exactly what an auditor would read as a genuinely new and independent problem.

A new finding is therefore not automatically independent evidence of a new underlying problem, because a prior finding may have changed how broadly the auditor searched. The converse holds with equal force: broadly elevated findings are also consistent with programs that are genuinely worse across the board, and with stable differences between auditees that make some of them durably more prone to deficiencies (Appendix A). The table narrows the space of explanations; it does not identify the mechanism. In the Clearinghouse data the elevation runs across repeat-only, mixed and any-new outcomes, which is the pattern the last two rows of Table 3 produce and the pattern a worse program produces, and the record cannot say which.

The design question that follows is not how to squeeze more out of the finding record but what additional information would separate the two factors in equation (2). Section 7 shows what happens when that question is skipped; Section 8 answers it.

7 RISK MODELS TRAINED ON FINDING HISTORIES

Audit organizations increasingly build models that score programs for risk and validate them the way any predictive model is validated: fit on one period, test on another, report an AUC, check calibration. The metrics are not broken. They answer a precise question about the label they are given, and the label is the recorded finding. The error is to carry the result over and describe the model as predicting where problems are. It has teeth because a score that steers where auditors look can make itself right: a flagged program gets more attention, more attention raises detection, higher detection produces recorded findings, and the recorded findings validate the score (Ensign et al., 2018; Lum and Isaac, 2016; Lakkaraju et al., 2017; Kleinberg et al., 2018). The obvious objection is that this is just a story about auditing more, so the construction worth making holds average detection fixed.

Figure 3 does that. Across the whole horizontal axis the average chance of catching an existing condition stays at 0.40 and the overall recorded finding rate stays at 0.08. The only thing that changes is allocation: detection moves away from the programs the score does not flag and toward the programs it does. The score is statistically independent of where the conditions actually are; it knows nothing. Measured against the recorded finding, its AUC climbs from 0.50, a coin flip, to 0.70, and its Brier score improves. Measured against the underlying condition, its AUC is exactly 0.50 at every point and its Brier score worsens by exactly the amount the recorded-label Brier score improves. On a validation report this reads as a model that became genuinely discriminating, with no change in average detection. The calculation is exact, from a single prespecified configuration. Appendix B gives the grid, the general result, a fitted rather than an oracle score, and the finite-sample distributions at portfolio sizes an audit organization actually has. Calibration is no stronger a guarantee: a model can be perfectly calibrated for recorded findings while carrying no information about the underlying condition at any point of its range, and the appendix works a case in which predictions span a factor of 94 across deciles while the underlying problem rate is flat in all ten.

The implication is narrow. Reporting that a risk model achieves a given AUC against recorded findings is a true and useful statement and should be made in exactly those words. An ordinary train/test split does not detect the problem, because both halves were produced by the same audit process under the same targeting policy. What cannot be concluded is that the model has located where problems are; if the model also influences audit planning, the validation evidence is partly a record of the model's own effect on the audit.

8 MAKING THE BOUND OPERATIONAL: LEARNING ENOUGH ABOUT DETECTION TO USE IT

The bound in equation (7) does work only when the organization can say something about M . Two things need to be clear about M before any route to it is described.

The first is what M is not. $M = d_1/d_0$ is the ratio of two probabilities: that an existing condition is detected after a prior finding, and that it is detected after a clean history. It is not audit effort. Hours, sample sizes, coverage percentages and procedure counts are not M , and doubling any of them does not double M ; they inform M only through a defensible connection to detection.

The second is that M is target-specific. The multiplier is defined for a particular detection target: a defined condition, finding class, procedure or audit objective. Different compliance requirements, finding types, control deficiencies and transaction tests can have different d_0 , different d_1 and therefore different M . An organization should not assume that follow-up has the same detection effect for every condition, requirement or procedure. In practice the useful object is M for a stated finding class, or a defensible bound that applies to a specified family of conditions; the text keeps the single symbol M for readability, with that scope understood.

Six ways to learn or bound M follow, each with different information requirements and identifying assumptions. Table 4 summarizes what each supplies and the assumption on which it turns.

Reference observation under a common measurement rule

The cleanest design is reference observation under a common measurement rule: a set of program-years in each history group, examined with the same reference procedure, chosen so that the reference observations represent the population whose condition rate is being estimated. The reference sample does not have to ignore prior history. It may be stratified on prior-finding status; what matters is that within each history group it represents that group, through probability sampling or known weights. Tax administrations maintain random examination programs beside operationally selected ones for this reason (Slemrod, 2019), and safety regulators have randomized inspection allocation (Levine et al., 2012).

Let s be the sensitivity of the reference procedure for the target condition, the probability it detects that condition when present. One extra symbol is needed because the reference procedure is a different procedure from the targeted audit, with its own detection probability, and s is unknown.

If s is the same in both history groups, the reference finding rates are $q_0^{\text{ref}} = r_0 s$ and $q_1^{\text{ref}} = r_1 s$, so

$$\frac{q_1^{\text{ref}}}{q_0^{\text{ref}}} = \frac{r_1 s}{r_0 s} = R_{\text{lat}}. \quad (11)$$

The reference procedure does not need perfect detection to recover the underlying risk ratio. It needs the same sensitivity in both history groups, because a common sensitivity cancels, and only then.

A reference procedure does not need perfect detection to identify the underlying risk ratio. It needs comparable detection in the two history groups, and negligible false positives.

With R_{lat} in hand, equation (3) gives the multiplier directly:

$$M = R_{\text{obs}}/R_{\text{lat}}. \quad (12)$$

In words: the targeted audit gives the ratio of finding rates between the two history groups. The reference work gives the ratio of underlying condition rates, under a common measurement rule. Their ratio identifies the multiplicative detection component M . If the targeted record shows the fourfold ratio of Table 1 and the reference sample shows findings twice as often after a prior finding as after a clean year, then $R_{\text{lat}} = 2$ and $M = 4/2 = 2$: the observed fourfold ratio factorizes into a twofold underlying-condition ratio and a twofold detection multiplier. Section 5 could not do this because the Clearinghouse contains no such sample.

Five assumptions carry the result (Proposition A.6). First, within each prior-history group the reference sample represents the population whose condition rate is being estimated, directly or through known sampling weights. Second, the same reference measurement rule is applied in both groups. Third, the reference procedure has common sensitivity across the two groups for the target condition. Fourth, the reference observation does not alter the condition before it is measured. Fifth, the reference procedure has negligible false positives for the target condition, or its false-positive process is separately modeled; a reference that records findings where no condition exists

does not cancel in the ratio. The third is the load-bearing one, and applying the same nominal procedure does not by itself guarantee it. Sensitivity can differ between the groups if the conditions in one group are more severe or of different types, if records are harder to obtain, if the procedure is implemented differently, or if the case mix differs; then the reference ratio $R_{\text{ref}} = q_1^{\text{ref}}/q_0^{\text{ref}}$ equals R_{lat} times s_1/s_0 and the cancellation fails. Exact equality is not needed for a usable result. If the organization can defend $\ell \leq s_1/s_0 \leq u$, then

$$R_{\text{lat}} \in [R_{\text{ref}}/u, R_{\text{ref}}/\ell], \quad (13)$$

to be intersected with the record-based set (8). A reference ratio of 2 with sensitivities defended to within ten percent of each other ($\ell = 0.9$, $u = 1.1$) places R_{lat} between 1.82 and 2.22. Reference-sample size should be planned from the desired precision and the expected condition or finding rates in the two history groups.

High-sensitivity reference procedures

A second anchor comes from a procedure whose sensitivity s for a narrowly defined condition is known, or can be bounded from below. Applied to a whole subpopulation with no false positives, its finding rate q^{ref} determines that subpopulation's condition rate in one of three grades:

- perfect sensitivity, $s = 1$: $r = q^{\text{ref}}$;
- known sensitivity, $0 < s < 1$: $r = q^{\text{ref}}/s$;
- defended lower bound, $s \geq s_L$: $r \in [q^{\text{ref}}, \min(1, q^{\text{ref}}/s_L)]$.

Near-perfect sensitivity is the third grade with s_L close to one: at $s_L = 0.95$, r is at most about 5.3 percent above q^{ref} . The targeted procedure's finding rate q in the same group then gives its detection probability, $d = q/r$, exactly in the first two grades and within $[q/\min(1, q^{\text{ref}}/s_L), q/q^{\text{ref}}]$ in the third. Anchors in both history groups yield d_0 , d_1 and M , or bounds on them (Proposition A.6).

The anchor rests on the procedure’s sensitivity, not on its coverage. Exhaustive coverage by itself does not make detection perfect. It removes sampling or coverage uncertainty, and it anchors the underlying rate only to the extent that the procedure’s sensitivity for the target condition is known or bounded. Where auditors speak of “certainty testing”, the certainty refers to inclusion in the tested population, not to certainty that every existing condition will be recognized. Testing every transaction for an exact duplicate payment can have very high sensitivity for that narrowly defined condition; testing every file for a judgment-dependent compliance failure need not, and the anchor is then only as good as the sensitivity that can be defended.

Independent reference adjudication

A representative or appropriately weighted sample of program-years within each history group can be re-adjudicated by a reviewer who does not see the original finding history and applies a fixed program of work. Blindness removes the history’s influence on the reviewer; it does not by itself create ground truth, and two cases have to be kept apart.

In the first, a sufficiently accurate reference or adjudication procedure is available for the target condition. Then the reference review approximates latent status, d_0 and d_1 are estimated directly as the share of reference-identified conditions the original audit had recorded in each history group, and M is their ratio; this is the audit analogue of correcting a diagnostic study for differential verification (de Groot et al., 2011). In the second, the reference review is itself imperfect. Then its own error process has to be addressed, through multiple independent procedures, a model of detection that treats the reference as one more imperfect measurement, or an external benchmark with known properties. Ordinary blind re-performance of the same work does not automatically solve the problem, because both reviews may miss the same condition for the same reason, and the shared misses are invisible to both.

Two distinct detection procedures

Two imperfect procedures can reveal how much each is missing. Suppose procedure A vouches transaction-level evidence and procedure B obtains an independent external confirmation or performs a genuinely different control-based test, both applied to the same programs. Under conditional independence of the two procedures' failures and stable sensitivity, and with no false positives, the share of the conditions B finds that A also finds identifies A's detection probability; the share of the conditions A finds that B also finds identifies B's; and the overlap then says how many conditions both missed. This is the logic ecologists use to estimate how often a species is present at sites where a survey did not see it (MacKenzie et al., 2002), and it needs no reference sample (Proposition A.7).

Two requirements carry it. The procedures' failures must be unrelated once the true condition is fixed, and each procedure's sensitivity must be stable across the program-years being pooled. Using two operationally different procedures is not enough on its own: if both tend to miss the same difficult cases, for example where documentation does not exist, their agreement looks reassuring while saying little about what both missed. Vouching paired with an external confirmation can satisfy the first requirement; two reviewers re-performing the same sample with the same method do not.

Randomized or quasi-random changes in audit attention

Some changes in how hard an audit looks arrive for reasons unrelated to the underlying condition: randomized rotation, random reference selections, externally imposed coverage rules, staffing changes plausibly unrelated to the control environment, randomized review depth. These reveal how finding rates change when observation changes. A change in audit attention is informative about detection only if the change does not independently alter, or select on, the underlying condition; that is the exclusion restriction, and a staffing increase introduced because controls deteriorated fails it. And such variation identifies relative detection effects between the regimes it creates, or tightens bounds; absolute condition prevalence requires an additional sensitivity anchor of the

kind described above.

Procedure-based bounds

An organization that can do none of the above may still be able to defend an upper bound $\overline{M}$ from what it knows about its own procedures: the fraction of the population covered, sample sizes, the number and type of compliance requirements tested, the use of targeted versus broad procedures, the strata tested with a high-sensitivity procedure, historical quality-review results, and any external validation study. These quantities are not M . They inform M only through an explicit model connecting audit procedures to detection probability, for instance a statement of what share of existing conditions a procedure of a given depth is expected to catch, and the bound is only as defensible as that model.

The arithmetic once $\overline{M}$ is in hand is simple. If the organization can defend $M \leq 1.5$ for the finding class in question and observes $R_{\text{obs}} = 3$, equation (7) gives $R_{\text{lat}} \geq 3/1.5 = 2$: even after giving follow-up the largest detection advantage the organization considers defensible, the underlying condition must still be at least twice as common in the prior-finding group, and with monotone detection the underlying ratio lies between 2 and 3. What the record could only place between q_1 and $1/q_0$ has become a range an audit committee can act on.

The remaining route is a structural model whose restrictions on how the condition evolves and how detection responds are defended on audit grounds; identification then has to be established for that specific model, and this paper does not take that route.

9 CONCLUSION AND BOUNDARIES

The paper set out to answer four questions.

What can the finding record tell us? A finding history predicts the future finding record. It fixes a floor under the underlying condition rate in every history group and, if follow-up can only raise detection, a ceiling on the underlying-condition risk ratio: the observed finding ratio. Where

the observed ratio is below one, the underlying association is negative; in data, that conclusion requires an upper confidence limit for the observed ratio below one.

What can it only bound? The underlying-condition risk ratio itself. Without information about detection, the record places it anywhere between q_1 and $1/q_0$, an interval that at ordinary finding rates contains one and runs well below it. A longer record does not narrow the interval, because every added year is another observation of the product $q = rd$.

Why is follow-up different from discovery? A prior finding is routing information: it says where to return. The follow-up procedures, not the prior finding, produce the evidence about whether the known condition remains. Finding histories can contain genuine information about both recurrence and other problems. But when the history also changes future observation, the record alone cannot determine how much of that relationship belongs to the underlying condition and how much belongs to the observation process. Follow-up uses history to revisit known conditions. Discovery requires observation whose measurement rule is not itself distorted by the history being evaluated, or a defensible model of that distortion. In Clearinghouse data, among programs audited as major programs in both years, prior findings predict recurrence and materially more findings not classified as repeats, and the record does not identify why.

What additional audit information would let us learn more? Anything that pins down the follow-up detection multiplier M for the finding class in question: reference observation under a common measurement rule, which recovers the underlying ratio and hence M when the reference procedure's sensitivity is common to both history groups or its ratio is bounded; a reference procedure of known or bounded sensitivity that fixes or bounds the condition rate; reference adjudication accurate enough to stand in for latent status; two procedures with independent failures; quasi-random changes in attention; or, at minimum, a defended ceiling $\bar{M}$, under which the underlying ratio lies in $[\max\{q_1, R_{\text{obs}}/\bar{M}\}, R_{\text{obs}}]$. Each is an addition to audit design, not a statistical method applied to the record.

Five boundaries travel with these conclusions. Repeat findings are not artifacts; a repeat finding in these data is a real reported deficiency the auditor identified as a recurrence. A “new” finding

is one the auditor did not classify as a repeat, a statement about that classification and not about when the problem arose or whether it was recorded before. The identity $q = rd$ and every bound built on it assume no false positives, or a modeled restriction on them. The predictive results are constructions showing what can happen when a score influences observation; whether a given deployed model has this property is an empirical question that in most cases nobody has checked. And no result in Section 5 identifies the mechanism behind the associations it reports. Nothing here recommends a particular protocol for a particular office.

The missing quantity is detection. Once an organization measures, bounds, or standardizes it, the finding record becomes much more informative about the underlying condition.

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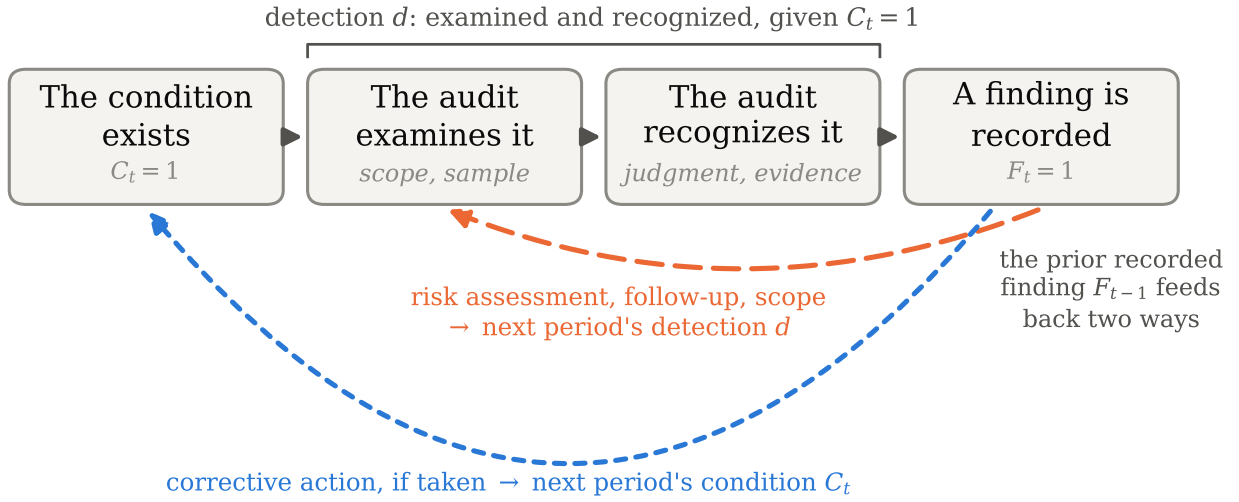


Figure 1: **A finding history is also an observation history.** A recorded finding ($F_t = 1$) requires the condition to exist ($C_t = 1$), the audit to examine the area where it is, and the auditor to recognize it there. The detection probability d covers both steps together, given that the condition exists: selection into observation as well as recognition. The absence of a recorded finding is therefore three different things wearing the same clothes: the condition was not there, or it was never looked for, or it was looked for and missed. The two feedback arrows are what make the record hard to read. The prior recorded finding F_{t-1} feeds back two ways: through corrective action, if any is taken, into next period's condition C_t , and through next period's risk assessment, follow-up plan and scope into next period's detection probability d . Equation (1) is this diagram written as a product, with F_{t-1} inside both factors. In the Single Audit the feedback is written into the regulation rather than inferred from behavior (Section 4).

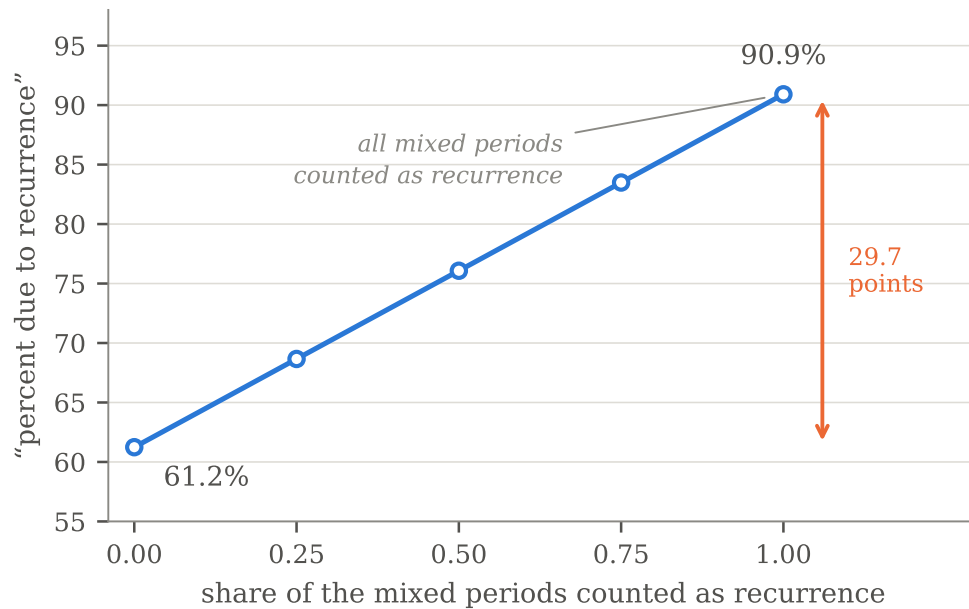


Figure 2: **A summary statistic that moves 29.7 points without the data moving at all.** Every program-period that recorded both a repeat finding and a new one has to be assigned somewhere before a “percent due to recurrence” figure can be computed, and those mixed periods are 25.9 percent of all prior-finding outcomes. Counting none of them as recurrence gives 61.2 percent; counting all of them gives 90.9 percent. Nothing observable differs between the two ends. The convention-free statement is the three-way split that the horizontal axis never touches: $0.5017 = 0.3073 + 0.0456 + 0.1488$, recurrence-only plus new-only plus both.

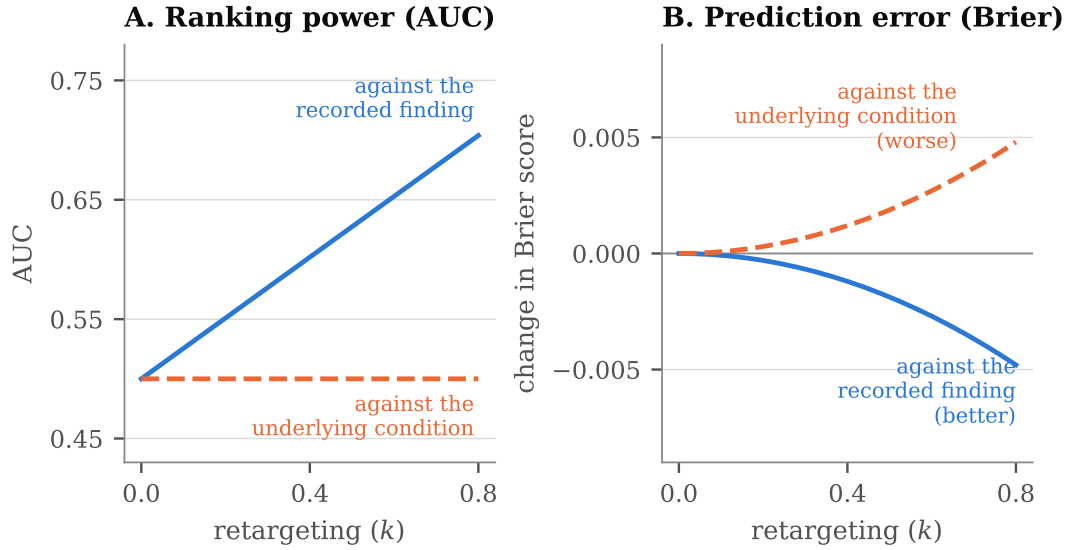


Figure 3: **Same average detection, stronger targeting.** Both panels hold average detection fixed: the average chance of catching an existing condition stays at 0.400 and the overall recorded finding rate stays at 0.080 at every point on the horizontal axis. That is a statement about detection, not about audit hours, staffing or cost. Only the *allocation* of detection changes, moving toward the programs the score flags, and the score is statistically independent of where the conditions are. Measured against the recorded finding, the score improves: AUC rises from 0.500 to 0.704, and the Brier score falls from 0.0736 to 0.0688. Measured against the underlying condition, the score is worth nothing at any point on the axis (AUC exactly 0.500), and its prediction error *rises*, from 0.1744 to 0.1792. Panel B plots both Brier series as the change from the untargeted starting point. Exact calculation; the grid is in Appendix B, and the finite-sample version is Figure 4 there. Solid series are measured against the recorded finding, dashed series against the underlying condition.

Table 1: **Three programs that produce exactly the same finding record.** Read down each column. In all three, a program with no finding last year has a 5 percent chance of a recorded finding this year and a program with a finding last year has a 20 percent chance, so the observed finding risk ratio R_{obs} is 4 in every column. Underneath, the first program’s condition persisted, the second program’s condition was no more likely than before and only the looking changed, and in the third the underlying rate is stipulated to be *half* as high after a finding year while detection is eight times as high. In each column the observed ratio is the product of the underlying-condition ratio R_{lat} and the history-conditioned detection multiplier M : $4 = 4 \times 1$, $4 = 1 \times 4$, $4 = 0.5 \times 8$. Exact arithmetic, not simulation.

	Persistence	Follow-up only	Remediation
<i>Chance the condition is there, $r \dots$</i>			
after a clean year, r_0	0.100	0.200	0.400
after a finding year, r_1	0.400	0.200	0.200
<i>Chance the audit detects it if it is there, $d \dots$</i>			
after a clean year, d_0	0.500	0.250	0.125
after a finding year, d_1	0.500	1.000	1.000
<i>What the record shows, $q = r \times d \dots$</i>			
after a clean year, q_0	0.05	0.05	0.05
after a finding year, q_1	0.20	0.20	0.20
Observed finding risk ratio, $R_{\text{obs}} = q_1/q_0$	4.0	4.0	4.0
Underlying-condition risk ratio, $R_{\text{lat}} = r_1/r_0$	4.0	1.0	0.5
Detection multiplier, $M = d_1/d_0$	1.0	4.0	8.0

A Identification

Each proposition is followed by a one-sentence statement of what it means in audit terms.

A.1 Setup

Fix a program and a period t . Let $C_t \in \{0, 1\}$ indicate that the underlying condition is present, and $F_t \in \{0, 1\}$ that a finding is recorded. Assume no false positives, so $F_t = 1$ implies $C_t = 1$. Write

Table 2: **What follows a prior-year finding.** Sample A: 206,926 program-years in which the same program was audited as a major program in both years, 39,067 with a finding in the prior year and 167,859 without. The four outcomes in the first panel are mutually exclusive and, under the auditor’s repeat flag used here, exhaust the possibilities, so their risk differences add to the difference in “any finding” exactly. Intervals are 95 percent cluster bootstrap, 1,000 replications resampling 45,952 auditees with their full histories, seed 20260918. Every interval excludes zero. The row that is easy to miss is “any new finding”: counting only the periods whose *sole* outcome was new understates it by roughly a factor of 2.23 on the ratio scale, because “new only” excludes every period that also recorded a repeat. A period can land in “both” because repeat follow-up is thorough, or because the program has several distinct deficiencies; these data do not separate the two. The ratio column is left blank for the three repeat-involving rows: with no prior finding a recorded repeat is close to impossible, so the baseline is structurally near zero and the ratio is arbitrarily large rather than informative. The risk differences are the usable quantities there.

Outcome for the program-year	Chance this year		Difference	95% interval	Ratio
	no prior finding	prior finding			
No finding at all	0.9234	0.4217	−0.5017	[−0.5104, −0.4927]	0.46
A repeat finding only	0.0015	0.3088	+0.3073	[0.3008, 0.3140]	—
A new finding only	0.0744	0.1200	+0.0456	[0.0416, 0.0498]	1.61
Both a repeat and a new finding	0.0007	0.1495	+0.1488	[0.1416, 0.1561]	—
Any finding	0.0766	0.5783	+0.5017	[0.4927, 0.5104]	7.55
Any repeat finding	0.0022	0.4583	+0.4561	[0.4458, 0.4656]	—
Any new finding (new only + both)	0.0751	0.2696	+0.1945	[0.1870, 0.2021]	3.59
<i>Findings counted rather than programs flagged — this needs no allocation rule:</i>					
Findings recorded, total	0.1029	1.1712	+1.0683	[1.0209, 1.1182]	
of which repeat	0.0027	0.7468	+0.7441	[0.7084, 0.7807]	
of which new	0.1002	0.4245	+0.3243	[0.3076, 0.3422]	

Table 3: How you look decides what turns up. Three ways of letting last year’s finding change this year’s audit, with the underlying problems held identical and drawn independently each period. Each cell is the change in the gap between programs with and without a prior finding, measured against a baseline in which prior findings change nothing. Following up the same finding moves repeats and *only* repeats. Looking harder across a requirement area also surfaces new findings inside that area. Looking harder at the program overall also surfaces findings in areas that never had one — which is what an auditor would otherwise read as a genuinely new problem. The zeros are exact. First-period histories, and with them the history groups (104,216 programs with a prior finding, 895,784 without) and each program’s set of previously flagged conditions and areas, are drawn once and shared by all regimes; second-period conditions and detection draws are shared unit by unit; and a regime changes detection only for the conditions it targets. An outcome outside a regime’s target set is therefore identical, program by program, to its baseline value. Simulation, 10^6 programs; Appendix C gives the full specification.

If last year’s finding makes the auditor...	Change in the prior-finding gap		
	A finding repeats	Something new, in an area that already had one	Something new, in an area that did not
Follow up the same finding	+0.0394	+0.0000	+0.0000
Look harder at the same requirement area	+0.0279	+0.0454	+0.0000
Look harder at the program overall	+0.0223	+0.0366	+0.1388

$f \in \{0, 1\}$ for the prior-period record F_{t-1} , and define

$$r_f = P(C_t = 1 \mid F_{t-1} = f), \quad d_f = P(F_t = 1 \mid C_t = 1, F_{t-1} = f),$$

$$q_f = P(F_t = 1 \mid F_{t-1} = f).$$

Call r_f the underlying rate, d_f the detection probability and q_f the recorded rate. Write $R_{\text{lat}} = r_1/r_0$, $R_{\text{obs}} = q_1/q_0$ and $M = d_1/d_0$, the history-conditioned detection multiplier. Every quantity conditions on the prior recorded history F_{t-1} : each compares groups defined by that history, and none is a causal effect of remediation or follow-up. The observable data determine q_0 and q_1 and nothing else. In Propositions A.1–A.4 take $q_0 > 0$ and $q_1 > 0$, which is the empirically relevant case and is needed for the ratios to be defined.

Proposition A.1 (Two-factor identity). $q_f = r_f d_f$ for $f \in \{0, 1\}$, and consequently $r_f \geq q_f$ and $R_{\text{obs}} = R_{\text{lat}} M$.

Table 4: **Six ways to learn or bound the follow-up detection multiplier.** What each design supplies and the assumption on which it turns. $M = d_1/d_0$ is defined for a stated condition, finding class or procedure; the designs are not ranked, because they answer different questions with different information requirements. Details and the remaining assumptions for each are in Section 8; the first two identifying results are Propositions A.6 and A.7 in Appendix A.

Design	What it supplies	Critical assumption
Reference observation under a common measurement rule	R_{lat} , hence $M = R_{\text{obs}}/R_{\text{lat}}$; with $s_1/s_0 \in [\ell, u]$, $R_{\text{lat}} \in [R_{\text{ref}}/u, R_{\text{ref}}/\ell]$	Common sensitivity in the two history groups, or a defended range for its ratio; no false positives
Reference procedure of known or bounded sensitivity	$r = q^{\text{ref}}$ if $s = 1$; $r = q^{\text{ref}}/s$ if s is known; $r \in [q^{\text{ref}}, \min(1, q^{\text{ref}}/s_L)]$ if $s \geq s_L$; hence $d = q/r$ or bounds on it	Known sensitivity, or a defended lower bound s_L , for the target condition; no false positives
Independent reference adjudication	d_0, d_1 directly	Reference status credible enough to stand in for the condition
Two distinct procedures	r, d within a history group	Conditionally independent failures; stable sensitivity; no false positives
Randomized or quasi-random attention shift	Relative detection; tighter bounds	Exclusion restriction: the shift does not alter or select on the condition
Procedure-based bound	$\overline{M}$, hence $R_{\text{lat}} \in [\max\{q_1, R_{\text{obs}}/\overline{M}\}, R_{\text{obs}}]$	A defensible model linking procedures to detection probability

Proof. $P(F_t = 1 \mid F_{t-1} = f) = P(C_t = 1 \mid F_{t-1} = f)P(F_t = 1 \mid C_t = 1, F_{t-1} = f) + P(C_t = 0 \mid F_{t-1} = f) \cdot 0 = r_f d_f$. Since $d_f \leq 1$, $r_f = q_f/d_f \geq q_f$. Dividing $q_1 = r_1 d_1$ by $q_0 = r_0 d_0$ gives the product form. $\square$

In audit terms: the recorded finding rate is the problem rate times the catch rate, so the problem rate is at least the recorded rate, and the recorded risk ratio is the underlying risk ratio times the detection multiplier.

Proposition A.2 (Unrestricted bounds). $R_{\text{lat}} \in [q_1, 1/q_0]$, and both endpoints are attained.

Proof. By Proposition A.1, $r_1 \in [q_1, 1]$ and $r_0 \in [q_0, 1]$, and these are the only restrictions the data impose: for any such pair, setting $d_f = q_f/r_f \in (0, 1]$ reproduces q_0 and q_1 exactly. Hence $R_{\text{lat}} = r_1/r_0$ is minimized at $r_1 = q_1$, $r_0 = 1$, giving q_1 , and maximized at $r_1 = 1$, $r_0 = q_0$, giving $1/q_0$. Both configurations are admissible, so both endpoints are attained. $\square$

On the difference scale the same argument gives $r_1 - r_0 \in [q_1 - 1, 1 - q_0]$, also sharp. The interval contains 1 (and the difference interval contains zero) whenever $q_1 < 1$ and $q_0 < 1$. In audit terms: the record tells us at least how common the problem must be in each group, but not how much more common it is in one history group than the other unless we also know something about detection.

Proposition A.3 (Monotone detection). If $d_1 \geq d_0$, then $R_{\text{lat}} \in [q_1, R_{\text{obs}}]$. Consequently:

- (i) $R_{\text{obs}} < 1 \implies R_{\text{lat}} < 1$;
- (ii) if in addition $d_1/d_0 \leq \bar{M}$ for a known $\bar{M} \geq 1$, then $R_{\text{lat}} \geq R_{\text{obs}}/\bar{M}$, so $R_{\text{obs}} > \bar{M} \implies R_{\text{lat}} > 1$;
- (iii) under $1 \leq d_1/d_0 \leq \bar{M}$ and no other restriction, the identified set for M is $[1, \min\{\bar{M}, 1/q_0\}]$ and the identified set for R_{lat} is $[\max\{q_1, R_{\text{obs}}/\bar{M}\}, R_{\text{obs}}]$; both are sharp.

Proof. From Proposition A.1, $R_{\text{obs}} = R_{\text{lat}}(d_1/d_0)$. If $d_1/d_0 \geq 1$ then $R_{\text{lat}} = R_{\text{obs}}/(d_1/d_0) \leq R_{\text{obs}}$; the lower bound is unchanged from Proposition A.2. Part (i) is immediate. For part (ii), $R_{\text{lat}} = R_{\text{obs}}/(d_1/d_0) \geq R_{\text{obs}}/\bar{M}$. For part (iii), $d_1 \leq 1$ and $d_0 = q_0/r_0 \geq q_0$ give $M \leq 1/q_0$, so every

admissible M lies in $[1, \min\{\bar{M}, 1/q_0\}]$. Conversely, for any M in that interval set $d_1 = 1$, $d_0 = 1/M$, $r_1 = q_1$ and $r_0 = q_0 M$; then $r_1 \in [q_1, 1]$, $r_0 \in [q_0, 1]$ because $1 \leq M \leq 1/q_0$, and $r_f d_f = q_f$. Every M in the interval is therefore attained, and $R_{\text{lat}} = R_{\text{obs}}/M$ traces $[R_{\text{obs}}/\min\{\bar{M}, 1/q_0\}, R_{\text{obs}}] = [\max\{q_1, R_{\text{obs}}/\bar{M}\}, R_{\text{obs}}]$, using $R_{\text{obs}} q_0 = q_1$. $\square$

In audit terms: if follow-up can only raise the catch rate, the recorded risk ratio is a ceiling on the underlying one, and a defended ceiling on how much follow-up raises the catch rate gives a floor on the underlying ratio. The Remediation column of Table 1 shows why the ceiling is needed: the recorded ratio is 4.0 and the underlying ratio is 0.5, reconciled by $M = 8$.

Proposition A.4 (Remediation reversal). Suppose $r_1 < r_0$, so the underlying condition is less likely after a recorded finding. Then $q_1 > q_0$ if and only if $d_1/d_0 > r_0/r_1$.

Proof. $q_1 > q_0 \iff r_1 d_1 > r_0 d_0 \iff d_1/d_0 > r_0/r_1$, the last step dividing by $r_1 d_0 > 0$. $\square$

In audit terms: when the problem has been reduced, recorded findings still go up after a prior finding whenever follow-up raises the catch rate by more than the problem was reduced; in the Remediation column, $r_0/r_1 = 2$ and $M = 8$.

Remark: dropping the no-false-positive assumption

Everything above assumes a recorded finding implies a real condition. Let $a_f = P(F_t = 1 \mid C_t = 0, F_{t-1} = f)$ be the false-positive rate after history f . Then

$$q_f = r_f d_f + (1 - r_f) a_f.$$

The multiplicative identity $q_f = r_f d_f$, the floor $r_f \geq q_f$ and every result built on them, Propositions A.1–A.5 and the design results below, hold when $a_0 = a_1 = 0$, or under a modeled restriction on (a_0, a_1) carried through the algebra. Without such a restriction the floor fails: $r_f = 0$ with $a_f = q_f$ reproduces any recorded rate, so the record alone does not bound r_f from below. The assumption is kept because it is the natural approximation for a formally adjudicated audit finding

and because it isolates the problem. Appendix B reports the corresponding predictive results with false positives allowed.

A.2 Full histories

Let h denote an observed history of recorded findings up to $t - 1$, of any length, and write $q(h) = P(F_t = 1 \mid h)$. The observable law of the process is exactly the collection $\{q(h)\}$ over histories h .

Proposition A.5 (Full-history nonidentification). Let $r(\cdot)$ be any function with $r(h) \in [q(h), 1]$ for every history h . There is a detection function $d(\cdot)$ taking values in $[0, 1]$ such that the model with underlying rates $r(\cdot)$ and detection $d(\cdot)$ reproduces $\{q(h)\}$ exactly.

Proof. For a history with $r(h) > 0$, set $d(h) = q(h)/r(h)$; this lies in $[0, 1]$ because $0 \leq q(h) \leq r(h)$. For a history with $r(h) = 0$, the constraint $r(h) \geq q(h)$ forces $q(h) = 0$, and $d(h)$ may be set to any value in $[0, 1]$, say 0. In both cases $r(h)d(h) = q(h)$, so by Proposition A.1 the model gives $P(F_t = 1 \mid h) = q(h)$ for every h . The observable law is determined by $\{q(h)\}$, which is unchanged. $\square$

In audit terms: however many years of findings are on file, any story about the underlying problem that stays above the recorded rate at every point can be made to fit the whole file exactly by adjusting the catch rate, so more years alone do not settle which story is true. The class of $r(\cdot)$ admitted includes functions increasing in the number of prior findings, constant functions, and decreasing functions.

Remark: what restricted latent-state models do recover

Proposition A.5 is not a statement that longitudinal data are uninformative about latent dynamics. Latent-structure models, hidden Markov models among them, are generically identifiable when enough periods are observed (Allman et al., 2009). Those results assume that the observed variables are conditionally independent given the latent state, together with homogeneity of the

transition law, a stable observation structure, and no additional unmodeled heterogeneity. History-dependent detection violates the conditional-independence assumption directly: F_t depends on F_{t-1} even after conditioning on C_t , because d is a function of F_{t-1} . That is the precise sense in which the identification problem here is not cured by more periods, and the precise sense in which a restricted model can cure it. If detection is restricted, fixed as a known function, tied to covariates observed outside the record, or measured directly, the model leaves the class covered by Proposition A.5. Leaving that class is necessary for identification and not sufficient for it: whether a particular restricted model is identified is a question about that model, to be settled for it specifically. What cannot work is leaving both factors free. Whenever the admissible class contains distinct underlying-rate functions satisfying $r(h) \geq q(h)$, the corresponding choices $d(h) = q(h)/r(h)$ (arbitrary where $r(h) = 0$) generate observationally equivalent models, and the class is not identified from the record alone.

A.3 Detection information from outside the record

The two results below formalize the two designs of Section 8 that identify M rather than bound it. Both are standard; they are stated here in the paper's notation.

Proposition A.6 (Reference observation under a common measurement rule). Suppose a reference procedure with sensitivity $s_f \in (0, 1]$ for the target condition in history group f and no false positives is applied to program-years that, within each history group f , represent that group, by probability sampling or through known weights, and that the procedure does not alter the condition. Let $q_f^{\text{ref}} > 0$ be the (weighted) reference finding rate after history f , and $R_{\text{ref}} = q_1^{\text{ref}}/q_0^{\text{ref}}$.

- (i) If $s_0 = s_1$, then $R_{\text{ref}} = R_{\text{lat}}$ and $M = R_{\text{obs}}/R_{\text{lat}}$.
- (ii) If $s_1/s_0 \in [\ell, u]$ with $0 < \ell \leq u$, then $R_{\text{lat}} \in [R_{\text{ref}}/u, R_{\text{ref}}/\ell]$, and intersecting with Proposition A.3(iii) gives the joint identified set.
- (iii) Within one group, $r_f = q_f^{\text{ref}}$ if $s_f = 1$; $r_f = q_f^{\text{ref}}/s_f$ if $s_f \in (0, 1)$ is known; and $r_f \in [q_f^{\text{ref}}, \min(1, q_f^{\text{ref}}/s_L)]$ if only $s_f \geq s_L > 0$ is defended.

Proof. Representativeness within each history group means the condition rate in the reference observations after history f is r_f , so $q_f^{\text{ref}} = r_f s_f$ by the argument of Proposition A.1. Hence $R_{\text{ref}} = R_{\text{lat}}(s_1/s_0)$, which gives (i), and (ii) follows by dividing by the two ends of $[\ell, u]$. For (iii), $r_f = q_f^{\text{ref}}/s_f$; with $s_f \in [s_L, 1]$ this lies in $[q_f^{\text{ref}}, q_f^{\text{ref}}/s_L]$, and $r_f \leq 1$. Then $M = R_{\text{obs}}/R_{\text{lat}}$ by Proposition A.1. $\square$

In audit terms: a reference sample that represents each history group, worked under the same rule in both, reveals how much more common the problem is after a prior finding even though the reference procedure misses some of it, because what it misses cancels in the ratio; dividing the targeted record's ratio by that number gives the detection multiplier. The sample may be stratified on history. If the procedure's sensitivity differs between the groups, the reference ratio equals $R_{\text{lat}} \cdot (s_1/s_0)$; a defended range for that ratio still bounds R_{lat} .

Proposition A.7 (Two conditionally independent procedures). Within one history group, apply two procedures A and B to the same program-years, with detection probabilities $a > 0$ and $b > 0$ given $C = 1$ that are constant across the program-years pooled, condition rate $r > 0$, no false positives, and $A \perp B \mid C$. Then P_A , P_B and P_{AB} below are positive. Let P_A , P_B and P_{AB} be the probabilities that A finds a condition, that B finds one, and that both do. Then $a = P_{AB}/P_B$, $b = P_{AB}/P_A$ and $r = P_A P_B / P_{AB}$.

Proof. $P_A = ra$, $P_B = rb$ and, by conditional independence, $P_{AB} = rab$. Then $P_{AB}/P_B = a$, $P_{AB}/P_A = b$, and $P_A P_B / P_{AB} = r$. $\square$

In audit terms: among the problems the second procedure finds, the share the first also found is the first procedure's catch rate, and once both catch rates are known the number of problems both missed follows; this is the occupancy-and-detection logic of MacKenzie et al. (2002), and it fails when the two procedures miss the same cases for the same reason, because then P_{AB} exceeds rab and both catch rates are overstated, or when sensitivity varies across the pooled program-years so that no single a or b exists. Applied in each history group it yields d_0 , d_1 and M for whichever procedure the targeted audit uses.

A.4 Serial dependence generated by observation alone

Proposition A.8 (Observation-only dynamics). Let $\{C_t\}$ be i.i.d. Bernoulli(p) with $0 < p < 1$, and generate the record sequentially: $F_t = C_t D_t$, where, given F_{t-1} , D_t is Bernoulli($d_{F_{t-1}}$) and independent of C_t and of all earlier (C_s, D_s) . Write $q_0 = p d_0$ and $q_1 = p d_1$, and assume $q_0 > 0$ and $q_1 < 1$. Initialize F_0 from the stationary distribution, $P(F_0 = 1) = \mu_*$ with

$$\mu_* = \frac{q_0}{1 - q_1 + q_0},$$

so that $0 < \mu_* < 1$. Then $\{F_t\}$ is a stationary two-state Markov chain with transition probabilities q_0 and q_1 and $P(F_t = 1) = \mu_*$ for every t ; its autocorrelation is $\rho_k = \lambda^k$ with $\lambda = q_1 - q_0 = p(d_1 - d_0)$; and $\rho_1 \leq p$.

Proof. Because C_t is independent of the past and D_t depends on the past only through F_{t-1} , $P(F_t = 1 \mid F_{t-1} = f, F_{t-2}, \dots) = p d_f = q_f$, so $\{F_t\}$ is Markov with transition probabilities q_0, q_1 . The value μ_* solves $\mu = \mu q_1 + (1 - \mu) q_0$, so initializing at μ_* makes the chain stationary, and $q_0 > 0, q_1 < 1$ give $0 < \mu_* < 1$, so $\text{Var}(F_t) = \mu_*(1 - \mu_*) > 0$. For a two-state chain, $P(F_{t+k} = 1 \mid F_t = f) = \mu_* + \lambda^k(f - \mu_*)$, hence $\text{Cov}(F_t, F_{t+k}) = \lambda^k \mu_*(1 - \mu_*)$ and $\rho_k = \lambda^k$. Finally $\lambda = p(d_1 - d_0) \leq p$ since $d_1 - d_0 \leq 1$. $\square$

In audit terms: even when the problem comes and goes at random with no memory at all, a follow-up rule makes the finding record look like it has memory, and the recorded risk ratio (M here, since $R_{\text{lat}} = 1$) says nothing about how strong that apparent memory is. In the exact case with $p = 0.10$, $d_0 = 0.05$ and $d_1 = 1.00$, the stationary finding prevalence is 0.0055, the recorded risk ratio is 20, and the lag-one autocorrelation of the record is 0.095. These are the exact stationary values; the accompanying simulation (Appendix C) starts every entity at $F = 0$ and discards 5 burn-in periods. Neither the magnitude of the recorded elevation nor the presence of serial dependence in the record discriminates between a persisting condition and a persisting audit response.

Remark: stable heterogeneity is a fourth channel

Persistence, observation feedback and remediation are not the only ways a finding record can look autocorrelated. Entities may simply differ: some auditees are durably more prone to deficiencies than others, and that alone produces association between one year's record and the next. Adding such heterogeneity to the multitype setting of Table 3 raises findings in requirement areas that had never had one, so it is a candidate explanation for the pattern that generalized scrutiny also produces. At the prespecified parameters it does not reproduce that pattern in magnitude: the raw new-area risk ratio is 0.718 with neither channel (below one, because of the opportunity-set artifact of Section 5), 1.02 with heterogeneity alone, and 2.06 with generalized scrutiny. This is a negative result at one set of parameters, not a demonstration that the two channels are distinguishable in general.

B Prediction against an endogenous label

This appendix holds the machinery behind Section 7: the general result, the binary and continuous constructions, the calibration table, the fixed-detection grid, the finite-sample distributions and the robustness checks. In audit terms, every result below says the same thing: a score that steers where the audit looks can validate well against the findings the audit then records while knowing nothing about where the conditions are.

B.1 Setup

Let S be a risk score, $C \in \{0, 1\}$ the underlying condition and $F \in \{0, 1\}$ the recorded finding, with no false positives. Let $P(C = 1) = p$ and let detection depend on the score, $d(S) = P(F = 1 \mid C = 1, S)$. Assume throughout that $S \perp C$: the score carries *no* information about where conditions are. Then $P(F = 1 \mid S) = pd(S)$, and any apparent performance of S against F is produced entirely by the dependence of detection on the score. The assumption is deliberately extreme; it isolates the mechanism. A real score with some genuine information inherits the same effect on top of

whatever genuine performance it has, and the two are indistinguishable from the validation output alone.

B.2 A binary score

Let $S \sim \text{Bernoulli}(\pi)$ independent of C , with detection d_1 when $S = 1$ and d_0 when $S = 0$. Write $\bar{d} = \pi d_1 + (1 - \pi) d_0$ and $\mu = P(F = 1) = p \bar{d}$, and assume $0 < \pi < 1$ and $0 < \mu < 1$, so that both score values and both labels occur.

Proposition B.1. $\text{AUC}(S, F) = \frac{1}{2} + \frac{\pi p (1 - \pi) (d_1 - d_0)}{2 \mu (1 - \mu)}$, while $\text{AUC}(S, C) = \frac{1}{2}$ exactly.

The second statement is immediate from $S \perp C$. At the prespecified values $p = 0.10$, $\pi = 0.25$, $d_0 = 0.10$ and $d_1 = 1.00$, the closed form gives $\text{AUC}(S, F) = 0.7683$; a Monte Carlo run at $N = 10^6$ returns 0.7685.

A labeling trap worth naming. The score in Proposition B.1 is a generic binary risk indicator drawn independently, not the lagged finding itself. If the score *is* the previous period's recorded finding, $S = F_{t-1}$, in the stationary regime of Proposition A.8, then

$$\text{AUC}(F_{t-1}, F_t) = \frac{1}{2} + \frac{1}{2} p (d_1 - d_0) \leq \frac{1 + p}{2},$$

which at $p = 0.10$ is bounded by 0.55, far below the 0.7683 available to a generic score. Within the observation-only process the lagged finding is a weak predictor of the record, because it is constrained by the same prevalence that limits the record. That is a statement about this model and not about lagged findings in general: where the underlying condition itself persists, the lagged finding carries information about that condition too, and the bound does not apply.

B.3 A continuous score

Let S be uniform on $[0, 1]$, independent of $C \sim \text{Bernoulli}(p)$ with $p = 0.20$, and let detection follow the logistic curve

$$d(S) = d_{\min} + \frac{d_{\max} - d_{\min}}{1 + e^{-\alpha(S-c)}}, \quad d_{\min} = 0.01, \quad d_{\max} = 0.99, \quad c = 0.80.$$

The steepness α takes the prespecified values 0, 2, 5, 10, 20 and 40, each simulated with $N = 10^6$ programs; exact values are computed by trapezoidal quadrature on a grid of 200,001 points. A sufficiently score-dependent detection process can produce high recorded-label AUC while the score remains independent of the underlying condition. The attainable AUC depends on the score distribution, the detection curve and average detection (Proposition B.2). In this configuration $\text{AUC}(S, F)$ rises with α ; at $\alpha = 40$ the exact value is 0.8918, with a Monte Carlo value of 0.8913, against a Monte Carlo $\text{AUC}(S, C) = 0.4995$. As α grows without bound, detection approaches a step at c and $\text{AUC}(S, F)$ approaches 0.897.

Proposition B.2 (AUC envelope). Let $S \perp C$ with $P(C = 1) = p \in (0, 1)$, let $d(S) \in [0, 1]$ with $\bar{d} = E[d(S)] \in (0, 1]$, and let $\mu = p\bar{d}$. Then

$$\text{AUC}(S, F) \leq 1 - \frac{(1-p)\bar{d}}{2(1-\mu)},$$

with ties counted one-half. Equality holds when detection is one on a set of scores of probability $\bar{d}$, every score in that set ranks above every score outside it, and detection is zero outside it.

Proof. Relative to the law of S , the law of S given $F = 1$ has density $d/\bar{d}$, and the law given $F = 0$ has density $(1 - pd)/(1 - \mu)$. Since $1 - pd = (1 - p)d + (1 - d)$, the second law is the mixture $\lambda L_1 + (1 - \lambda)G$, where L_1 is the law given $F = 1$, $\lambda = (1 - p)\bar{d}/(1 - \mu) \in (0, 1]$, and G has density $(1 - d)/(1 - \bar{d})$ (if $\bar{d} = 1$, $\lambda = 1$). AUC is linear in the law of the negative class, a law compared with itself has AUC one-half, and any AUC is at most one, so $\text{AUC}(S, F) \leq \lambda/2 + (1 - \lambda) = 1 - \lambda/2$. Equality requires $\text{AUC}(L_1, G) = 1$, which the stated configuration achieves. $\square$

In audit terms: however detection is steered, a score that knows nothing about the condition cannot exceed this ceiling against the record, and it can reach the ceiling. At $p = 0.20$ and $\bar{d} = 0.400$, the values of the fixed-average-detection construction below, the ceiling is 0.8261; that construction reaches 0.704 because its binary score flags only a quarter of programs and caps detection at one within them.

B.4 Calibration

Proposition B.3. Let $m(S) = pd(S)$. Then $E[F | m] = m$: the model is perfectly calibrated for the recorded finding. But $E[C | m] = p$ for every value of m : it carries no information about the underlying condition at any point of its range.

Proof. $E[F | S] = P(F = 1 | S) = pd(S) = m(S)$, and m is a function of S , so $E[F | m] = m$. Since $S \perp C$, $E[C | S] = p$, hence $E[C | m] = p$. $\square$

The calibration table uses the continuous configuration above at $\alpha = 20$.

Table 6 shows this numerically. Across deciles of the score, mean predictions run from 0.0020 to 0.1872, a span of roughly 94-fold, and the recorded finding rate tracks them closely, while the underlying problem rate sits between 0.198 and 0.202 in every decile. A calibration plot against an endogenously observed label is a statement about that label.

B.5 Fixed average detection

The constructions above can be dismissed as saying only that more attention produces more findings. This one cannot. The quantity held fixed is $\bar{d}$, the average probability of detecting an existing condition across the portfolio; it is not a measure of audit hours, staffing or cost. Fix $\bar{d}$ and set

$$d_0 = \bar{d} - \pi k, \quad d_1 = \bar{d} + (1 - \pi)k.$$

Then average detection $\pi d_1 + (1 - \pi)d_0 = \bar{d}$ is invariant in k , and so is the recorded prevalence $\mu = p\bar{d}$. The parameter k moves detection from unflagged cases to flagged ones without changing the total.

At the prespecified values $p = 0.20$, $\pi = 0.25$ and $\bar{d} = 0.400$, evaluated at 21 points on $k \in [0, 0.8]$, six quantities behave as the construction requires and were checked at every point. Average detection stays at 0.400 and recorded prevalence at 0.080. The AUC against the recorded finding rises monotonically from 0.500 to 0.704, with the closed form and the direct computation agreeing throughout. The AUC against the underlying condition is exactly 0.500 at every point. The Brier score against the recorded finding falls monotonically from 0.0736 to 0.0688, and the Brier score against the underlying condition rises monotonically from 0.1744 to 0.1792. Table 5 gives the grid. The configuration was fixed in advance and no parameter search was performed. The two Brier series move by exactly the same amount in opposite directions, for a reason that holds for every score.

Proposition B.4 (Brier identities). Let $S \perp C$, $m(S) = pd(S)$ and $\mu = p\bar{d}$. Then

$$\text{Brier}_F = E[(m - F)^2] = \mu(1 - \mu) - p^2 \text{Var} d(S),$$

$$\text{Brier}_C = E[(m - C)^2] = p(1 - p) + (p - \mu)^2 + p^2 \text{Var} d(S).$$

In the fixed-average-detection construction $\text{Var} d(S) = \pi(1 - \pi)k^2$, so $\Delta \text{Brier}_C = -\Delta \text{Brier}_F = p^2 \pi(1 - \pi)k^2$.

Proof. $E[F | S] = m$ gives $E[(m - F)^2] = E[m] - E[m^2] = \mu - \mu^2 - \text{Var} m$. $S \perp C$ gives $E[mC] = p\mu$, so $E[(m - C)^2] = E[m^2] - 2p\mu + p = \text{Var} m + (p - \mu)^2 + p(1 - p)$. Finally $\text{Var} m = p^2 \text{Var} d(S)$, and for the binary score $\text{Var} d(S) = \pi(1 - \pi)(d_1 - d_0)^2 = \pi(1 - \pi)k^2$. $\square$

At the end of the grid the shift is 0.0048 in each direction. Holding average detection fixed, targeting improves the recorded-label Brier score by exactly the amount it worsens the condition-label Brier score.

B.6 Robustness of the predictive results

Sample size. Figure 4 gives the sampling distributions at portfolio sizes an audit organization actually has. At 500 programs the middle 95 percent of $\text{AUC}(S, F)$ runs from 0.629 to 0.852, a range a reviewer would read as clear evidence of a working model, and $\text{AUC}(S, C)$ brackets one-half at every sample size tested, with 500 replications per configuration.

A fitted model rather than an oracle. The scores above are oracles: the exact functional form is known. Fitting a conventional flexible learner instead (gradient-boosted trees of depth three, learning rate 0.05, 200 iterations, five-fold stratified cross-fitting, on the continuous configuration at $\alpha = 10$) recovers the oracle closely and inherits its property: correlation with the oracle between 0.9994 and 0.9996, $\text{AUC}(S, F) = 0.8328$ against an oracle value of 0.8332, and $\text{AUC}(S, C) = 0.4996$. Nothing about the phenomenon requires the analyst to know the true detection function; an ordinary model discovers it from the data.

False positives. Relaxing the no-false-positive assumption, so that a finding can be recorded where no condition exists at rates a_0 and a_1 and the recorded rate is $pd_s + (1 - p)a_s$ for score value s , attenuates the recorded risk ratio but does not overturn the qualitative conclusion: with $a_0 = a_1 = 0.005$, $d_0 = 0.20$ and $d_1 = 0.80$, 11.5 percent of recorded findings are false positives, $\text{AUC}(S, F) = 0.6478$ and $\text{AUC}(S, C) = 0.5002$.

C Empirical detail

C.1 Data acquisition and panel construction

The Federal Audit Clearinghouse publishes the complete structured results of every Single Audit submission as bulk files at <https://www.fac.gov/data/download/>, without registration, authentication or fee. Seven files were retrieved in September 2026 and cover fiscal years 2016 onward. No free-text finding narratives, no restricted data and no nonpublic records of any kind

were used. Regulatory quotations are from the 2023 annual edition of Title 2 of the Code of Federal Regulations. The provisions the paper relies on, the audit threshold (200.501(a)), the tiered Type A threshold (200.518(b)(1)), the Type A low-risk criteria (200.518(c)) and the follow-up requirement (200.514(e)), applied with the same thresholds and criteria to every audit in the sample.

The unit of analysis is the auditee–program–year. A “program” is the regulatory program unit: a cluster of programs where one exists, otherwise a single assistance listing number. That follows 2 CFR 200.518(b)(3), under which “a cluster of programs is treated as one program” for the Type A determination, so the unit matches the one the rules operate on. The window is fiscal years 2016 through 2023, restricted to audits whose fiscal year began between 26 December 2014 and 30 September 2024. Over that period the audit threshold in 200.501(a) was \$750,000, and the Type A threshold in 200.518(b)(1) was tiered by the entity’s total federal expenditures, with \$750,000 as its lowest tier, applying to entities expending up to \$25 million. The 2024 revision raised the audit threshold and the lowest Type A tier to \$1,000,000 for fiscal years beginning on or after 1 October 2024 (Office of Management and Budget, 2024). The panel ends before that revision takes effect, so a single threshold regime governs throughout.

The panel is built from three bulk files. Audits (the general file) are limited to single audits; program-specific audits are excluded. Duplicate valid submissions, identified by the Clearing-house’s own rule (same auditee UEI, EIN, audit year, entity name and auditee e-mail), are collapsed to the most recently accepted record, with the report identifier breaking ties. The panel keeps EIN-years with exactly one accepted audit, so that EIN, program unit and audit year form a unique key; the build aborts if any key is duplicated. Award lines are joined to audits by report identifier and rolled up to the program unit, and a program unit is audited as major in a year when any of its award lines is marked major. Findings are joined to award lines by report identifier and award reference and rolled up to the program unit, each distinct finding reference counted once even when it is attached to several award lines. A program-year’s prior year is the same EIN and program unit in audit year $t - 1$.

Table 7 shows how the panel narrows and why a missing prior year is not coded as a clean one.

The panel distinguishes a resolved prior-year observation with no finding from the absence of any prior-year record for that program: of the 2,387,511 program-years, 66,131 have a resolved prior-year finding and 1,534,767 a resolved prior-year clean result; the remainder are excluded rather than recoded. Sample A restricts further to program-years in which the program was audited as major in both years, so every Sample A quantity is conditional on $J_{t-1} = J_t = 1$. That conditioning does not make selection ignorable, and it does not hold audit intensity or condition-specific observation fixed, and cannot, for the reason given in Section 4: the follow-up requirement applies within Sample A.

Two definitions of “repeat” are used. The primary one is the auditor’s own coding. The secondary one is a validated historical link: a finding whose cited prior reference number appears as a finding reference recorded for the same auditee and program in the immediately preceding year. Of the 67,157 findings that cite a prior-finding reference, 40,930 resolve this way, 60.9 percent. Non-resolution has innocent causes: the prior finding may sit two or more years back, may be recorded against a different program unit, or may carry a renumbered reference. Neither measure is a gold standard, so both are reported. Intervals throughout are 95 percent cluster bootstrap with 1,000 replications, resampling 45,952 auditees together with their full histories, seed 20260918.

C.2 Every specification

Table 8 collects the risk differences under all five specifications reported anywhere in this paper. The first row is the primary result of Table 2. The second substitutes the validated historical link for the auditor’s repeat flag. This definition mechanically forces both repeat states to zero for program-years with no prior finding, so its risk differences are interpretable and its risk ratios are not. It has a second consequence, which the table makes explicit. Under the auditor’s flag, every recorded finding is either marked a repeat or is not, so the four outcome states exhaust the possibilities and their risk differences add to the difference in “any finding”. Under the validated link they do not. A program-year whose findings were all flagged as repeats, none of which resolves to a same-program prior-year reference, has recorded no new finding and no *validated* repeat, and therefore

falls outside all four states. That fifth state occurs in 0.0015 of program-years with no prior finding and 0.0094 of those with one, a difference of 0.0079, which is exactly the gap between the four state columns and “any finding” in that row. It is given its own column rather than absorbed anywhere, and the sum is checked in every row.

The third row drops the requirement that the program be audited as major in both years, using all program-years with a resolved prior observation. The fourth restricts to transitions whose Clearinghouse provenance is stable across the two years, which addresses the possibility that repeat-flag capture differs between the two administering systems that operated during the window; that restriction excludes 39,909 transitions. The fifth uses the three-year cumulative history rule, requiring resolved observations at $t - 1$, $t - 2$ and $t - 3$. Across all five, the any-new margin is the most stable quantity in the table.

C.3 The allocation of mixed periods

Table 9 gives the full sensitivity of the “percent due to recurrence” statistic to the treatment of program-periods that recorded both a repeat and a new finding. ζ is the fraction of mixed periods counted as recurrence. The $\zeta = 1$ row gives 90.9 percent; the $\zeta = 0$ row gives 61.2 percent. The arithmetic is not in doubt: the numerator $0.3073 + \zeta \cdot 0.1488$ and the denominator 0.5017 are read directly off Table 2. For each fixed ζ the statistic is a well-defined descriptive function of observable quantities. The range across ζ shows that it is not uniquely defined until ζ is stated, and no value of ζ makes it a causal attribution.

C.4 Compliance-requirement categories

The Clearinghouse records a requirement category for every reported deficiency. It does not record which requirements applied to, or were tested in, a given program-period; that universe is defined by the Compliance Supplement, which is external to these data. A first-in-area hazard therefore has no constructible denominator here, and none was constructed.

What can be said is descriptive and applies only to programs that recorded findings in both

years. Among those 22,594 Sample A program-years, 86.4 percent share at least one requirement category across the two years; the mean Jaccard overlap between the two category sets is 0.664; on average 0.476 categories enter and 0.555 exit between the two years. These are descriptions of a conditioned subset, not comparisons across history groups, and they carry no implication about the rate at which problems appear in new requirement areas.

C.5 Horizons beyond one year

Results at $t + 2$ and $t + 3$ are not reported in the main text, and the reason is structural rather than statistical. Following a program past $t + 1$ requires it to be audited as a major program again, and that is exactly the quantity the regulation makes depend on the finding record: a prior finding can block low-risk Type A status and so make recoverage more likely. The programs that can be followed are therefore not a random subset of the programs one would want to follow. Without a model of recoverage, the direction of the resulting selection bias is not determined.

The attrition is severe. Of 21,795 Sample A finding-years with resolved follow-up at $t + 1$, 14,212 can be followed to $t + 2$ and 10,143 to $t + 3$, losses of 34.8 and 53.5 percent. Across those shrinking and increasingly selected samples the explicit-repeat rate falls from 0.4661 at $t + 1$ to 0.3004 at $t + 3$. That decline is consistent with decay of the condition, with selective recoverage, or with both, and the design cannot separate them, which is why the numbers are recorded here rather than interpreted.

C.6 Simulation specifications

Every simulation draws from a generator seeded by the master seed 20260918 and the experiment's name, so each experiment is reproducible on its own. The quantities reported are either exact calculations or Monte Carlo results from a single prespecified configuration; no parameter search was performed.

Table 1 and the remediation reversal. Exact arithmetic from the stated r_f and d_f ; no simulation.

Table 3. 10^6 programs, two periods, twelve condition types in four requirement areas of three types each. The per-type prevalences, by area, are $(0.08, 0.06, 0.04)$, $(0.07, 0.05, 0.03)$, $(0.06, 0.04, 0.02)$ and $(0.05, 0.03, 0.015)$. In period one each condition is present independently with its prevalence and recorded with probability 0.20. The prior-finding group is the programs with at least one period-one finding (104,216 programs; 895,784 have none). Period-two conditions are drawn afresh, independently of period one, with the same prevalences. Period-two detection is 0.20 except as the regime specifies: 0.90 for condition types recorded in period one (same finding); 0.70 for every type in a requirement area with a period-one finding (same area); 0.60 for every type in a program with any period-one finding (program overall). A period-two finding is a repeat if its type was recorded in period one, new in an old area if its type was not recorded but its area was, and new in a new area otherwise. Each outcome is the indicator that at least one finding of that kind is recorded; the gap is its rate in the prior-finding group minus its rate in the other group; each cell is the regime’s gap minus the baseline gap. Period-one draws, history groups, period-two conditions and period-two detection uniforms are shared by all regimes, and a regime changes detection only for the types it targets, so outcomes outside the target set are identical program by program. The heterogeneity check in Appendix A multiplies every prevalence by $\theta/E[\theta]$, with $\theta \sim \text{Beta}(2, 8)$ fixed for a program across both periods and prevalences capped at 0.80, and reruns the baseline, same-finding and program-overall regimes.

Observation-only dynamics (Proposition A.8). Reported values are exact stationary quantities. The accompanying simulation uses 250,000 entities over 20 periods, starts every entity at $F = 0$, and discards the first 5 periods.

Prediction (Appendix B). The binary score uses $N = 10^6$ with the closed form of Proposition B.1 as the check. The continuous score, the calibration table and the fitted model use the configuration stated in Appendix B. The fixed-average-detection grid is exact, with k at 21 equally spaced points from 0 to the largest value keeping $d_0 \geq 0$ and $d_1 \leq 1$. The finite-sample figure draws 500 portfolios at each size from 500 to 10,000 and 200 at 50,000. The false-positive check uses

$N = 10^6$.

C.7 Data and code availability

Reproduction requires the seven bulk files, a build of the auditee–program–year panel following the definitions above, and estimation of the outcome shares with cluster-bootstrap intervals as described. The panel construction, the four-state decomposition, the cluster bootstrap and the simulations reported in Sections 6 and 7 are implemented in Python and DuckDB with a fixed master seed. The quantities reported from simulation are either exact calculations or Monte Carlo results from a single prespecified configuration; no parameter search was performed, and each table states which of the two it is. The full analysis code, the intermediate panel and cryptographic digests of every input and output will be deposited with the journal on conditional acceptance and retained for at least seven years.

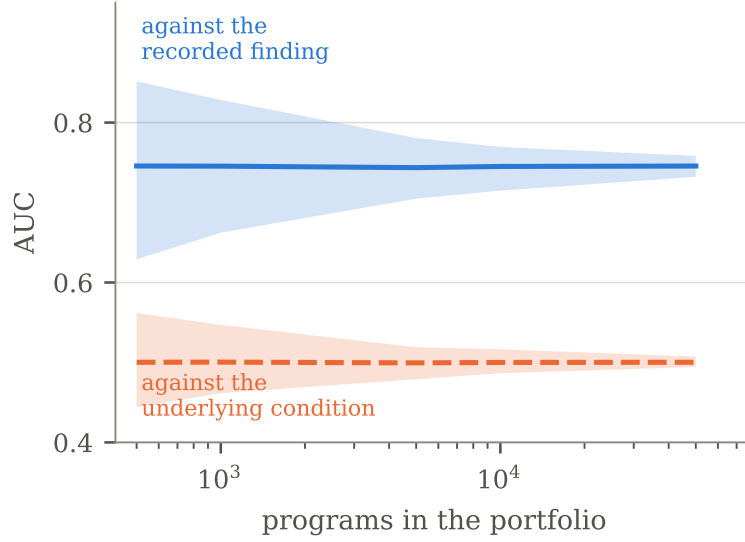


Figure 4: **Not a large-sample artifact.** Sampling distributions of the AUC of a score with no information about the underlying condition, measured against the recorded finding (solid) and against the condition (dashed), at portfolio sizes an audit organization actually has. Bands are the middle 95 percent across 500 replications per point (200 at 50,000 programs). At 500 programs the band against the record runs from 0.629 to 0.852, while the band against the condition brackets 0.5 at every size. Binary score: $p = 0.10$, $\pi = 0.25$, $d_0 = 0.10$, $d_1 = 0.80$, a separate prespecified configuration from Figure 3; simulation, seed 20260918.

Table 5: A fixed detection-capacity budget, every fourth grid point of 21. What is held fixed is $\bar{d}$, the average probability of detecting an existing deficiency; it is not a measure of audit hours or cost. Detection is set by $d_0 = \bar{d} - \pi k$ and $d_1 = \bar{d} + (1 - \pi)k$, with $p = 0.20$, $\pi = 0.25$ and $\bar{d} = 0.40$, all prespecified. Average detection and recorded prevalence are invariant in k by construction; the table shows they are. Exact calculation.

k	d_0	d_1	average detection	recorded rate	AUC vs F	AUC vs C	Brier vs F	Brier vs C
0.00	0.400	0.400	0.400	0.080	0.5000	0.5000	0.0736	0.1744
0.16	0.360	0.520	0.400	0.080	0.5408	0.5000	0.0734	0.1746
0.32	0.320	0.640	0.400	0.080	0.5815	0.5000	0.0728	0.1752
0.48	0.280	0.760	0.400	0.080	0.6223	0.5000	0.0719	0.1761
0.64	0.240	0.880	0.400	0.080	0.6630	0.5000	0.0705	0.1775
0.80	0.200	1.000	0.400	0.080	0.7038	0.5000	0.0688	0.1792

Table 6: Calibrated for the record, uninformative about the problem. Deciles of the oracle score $m(S) = pd(S)$ at $\alpha = 20$. Predicted values track the recorded finding rate across a span of roughly 94-fold, while the underlying problem rate is flat at $p = 0.20$ in every decile. Simulation, $N = 10^6$.

Decile of the score	Mean prediction	Recorded finding rate	Underlying problem rate
1	0.0020	0.0019	0.2004
2	0.0020	0.0022	0.1996
3	0.0020	0.0020	0.2006
4	0.0020	0.0022	0.2009
5	0.0022	0.0022	0.1983
6	0.0035	0.0036	0.2011
7	0.0127	0.0132	0.2024
8	0.0573	0.0566	0.1997
9	0.1417	0.1423	0.1998
10	0.1872	0.1852	0.1983

Table 7: How the sample was built, and what was left out. The upper panel narrows the panel to program-years in which the same program can be observed in two consecutive years with its major-program status known in both. The lower panel is the reason the paper does not simply code a missing prior year as a clean year. In 366,506 program-years — 15.35 percent of the panel — the auditee filed an audit last year but this program does not appear in it. Those program-years record a finding 3.26 percent of the time, which is not the rate of a clean program. They are excluded rather than recoded, because the absence of a record is not a clean result.

Step	Program-years	Excluded here
All program-years in the panel	2,387,511	
Within the \$750,000 regime (FY2016–2023)	2,387,511	0
With a resolved prior-year observation	1,600,898	786,613
Sample A: audited as a major program in both years	206,926	1,393,972

What the prior year looks like	Program-years	Share	Finding rate	
			this year	Treatment
Prior year resolved, finding recorded	66,131	2.77%	0.4723	kept
Prior year resolved, no finding recorded	1,534,767	64.28%	0.0206	kept
Auditee filed last year; this program is not in that filing	366,506	15.35%	0.0326	excluded
No filing at all for this auditee last year	160,861	6.74%	0.0706	excluded
First year of the window (2016)	259,246	10.86%	0.0462	excluded

Table 8: Risk differences under every specification reported. The primary row is the one in Table 2. The validated-link row mechanically forces the repeat states to zero for program-years with no prior finding, so its risk differences are usable and its ratios are not. It also opens a fifth outcome state: a period whose findings were all auditor-flagged repeats, none of which resolves to a same-program prior-year reference, records no new finding and no validated repeat, and so falls outside all four. That column is empty for every other specification, where the four states do exhaust the possibilities. In each row the four state columns sum to “any finding”, which the exhibit script asserts on the unrounded values; the columns here are rounded independently, so a row may add up one unit in the last place away from the total shown. The any-new margin is the quantity that moves least across all five specifications.

Specification	n	Mutually exclusive outcome states					
		repeat only	new only	both	flagged repeat, not linked	any finding	any new finding
Sample A, auditor flag (primary)	206,926	+0.3073	+0.0456	+0.1488	—	+0.5017	+0.1945
Sample A, validated link	206,926	+0.2994	+0.0491	+0.1454	+0.0079	+0.5017	+0.1945
All resolved transitions	1,600,898	+0.2869	+0.0617	+0.1031	—	+0.4517	+0.1648
Stable provenance	167,017	+0.2935	+0.0510	+0.1439	—	+0.4885	+0.1949
Three-year history	94,360	+0.1959	+0.0774	+0.0942	—	+0.3674	+0.1716

Table 9: The recurrence share as a function of a reporting convention. ζ is the fraction of mixed program-periods — those recording both a repeat and a new finding — counted as recurrence. Counting all of them gives 90.9 percent; counting none gives 61.2 percent. Nothing observable differs between the two ends, which is why the paper reports the three-way split instead.

ζ	“due to recurrence”	“first recorded”
0.00	61.2%	38.8%
0.25	68.7%	31.3%
0.50	76.1%	23.9%
0.75	83.5%	16.5%
1.00	90.9%	9.1%